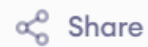


Main.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the value of N: ");
10
11         if (!sc.hasNextInt()) {
12             System.out.println("Invalid Input! Please enter
13                             an integer.");
14             sc.close();
15             return;
16         }
17
18         int n = sc.nextInt();
19
20         if (n <= 0) {
21             System.out.println("Invalid Input! N should be
22                             greater than 0.");
23             sc.close();
24             return;
25         }
26     }
27 }
```

```
Enter the value of N: 5
Enter 5 numbers:
1 2 3 4 5
Exception Caught: Array index is beyond the size of array.
Sum = 15
```

=== Code Execution Successful ===

Main.java



Share

Run

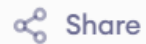
Output

Clear

```
1 import java.util.Scanner;
2
3 class FibonacciRunnable implements Runnable {
4
5     int n;
6
7     FibonacciRunnable(int n) {
8         this.n = n;
9     }
10
11     public void run() {
12
13         int first = 0;
14         int second = 1;
15
16         System.out.print("Fibonacci Series: ");
17
18         for (int i = 1; i <= n; i++) {
19
20             if (i == 1) {
21                 System.out.print(first + " ");
22             }
23             else if (i == 2) {
24                 System.out.print(second + " ");
25             }
26             else {
```

```
Enter number of terms: 5
Thread State: NEW
Fibonacci Series: 0 1 1 2 3
=== Code Execution Successful ===
```

Main.java



Share

Run

Output

Clear

```
1- import java.util.Scanner;
2
3- class MultiplicationThread extends Thread {
4
5     int number;
6
7-     MultiplicationThread(int number) {
8         this.number = number;
9     }
10
11-     public void run() {
12
13         System.out.println("Thread for " + number + " is
            Running");
14
15-         for (int i = 1; i <= 10; i++) {
16             System.out.println(number + " X " + i + " = " +
                (number * i));
17         }
18
19         System.out.println("Thread for " + number + " is
            Terminated");
20     }
21 }
22
23- public class Main {
```

```
Enter two numbers: 5 10
Thread 1 State: NEW
Thread 2 State: NEW
Thread for 10 is Running
Thread for 5 is Running
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
10 X 1 = 10
10 X 2 = 20
10 X 3 = 30
10 X 4 = 40
10 X 5 = 50
10 X 6 = 60
10 X 7 = 70
10 X 8 = 80
Thread for 5 is Terminated
10 X 9 = 90
10 X 10 = 100
```

Main.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 class SuperClass {
4
5     int value;
6
7     // Parameterized constructor of Super Class
8     SuperClass(int value) {
9         this.value = value;
10    }
11 }
12
13 class SubClass extends SuperClass {
14
15     int value; // Hides the member of Super Class
16
17     // Constructor of SubClass
18     SubClass(int superValue, int subValue) {
19
20         // Calling Super Class parameterized constructor
21         super(superValue);
22
23         this.value = subValue;
24     }
25
26     void display() {
```

Enter two values: 100 200

Super Class Value: 100

Sub Class Value: 200

=== Code Execution Successful ===

Bank.java



Share

Run

Output

```
1 import java.util.Scanner;
2
3 class Bank {
4
5     double getRateOfInterest() {
6         return 0;
7     }
8
9     public static void main(String[] args) {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter Bank Name: ");
14         String bankName = sc.next().toUpperCase();
15
16         Bank bank;
17
18         switch (bankName) {
19
20             case "SBI":
21                 bank = new SBI();
22                 break;
23
24             case "ICICI":
25                 bank = new ICICI();
26             .
27         }
```

Enter Bank Name: SBI

SBI Rate of Interest = 8.4%

=== Code Execution Successful ===

RemoveDuplicates.java



Share

Run

Output

Clear

```
14         sc.close();
15         return;
16     }
17
18     int[] arr = new int[n];
19
20     for (int i = 0; i < n; i++) {
21         System.out.print("Enter element" + (i + 1) + ":");
22         arr[i] = sc.nextInt();
23     }
24
25     ArrayList<Integer> unique = new ArrayList<>();
26
27     for (int i = 0; i < n; i++) {
28         if (!unique.contains(arr[i])) {
29             unique.add(arr[i]);
30         }
31     }
32
33     System.out.println("Non-duplicate items:");
34     System.out.println(unique);
35
36     sc.close();
37 }
38 }
```

```
Enter the number of elements in array: 7
Enter element1: 10
Enter element2: 0
Enter element3: 20
Enter element4: 20
Enter element5: 30
Enter element6: 30
Enter element7: 40
Non-duplicate items:
[10, 0, 20, 30, 40]
```

=== Code Execution Successful ===

SpecialCharacterCount.java



Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class SpecialCharacterCount {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the statement: ");
9         String str = sc.nextLine();
10
11         int count = 0;
12
13         for (int i = 0; i < str.length(); i++) {
14             char ch = str.charAt(i);
15
16             // Count special characters
17             if (!Character.isLetterOrDigit(ch) && !Character
                .isWhitespace(ch)) {
18                 count++;
19             }
20         }
21
22         System.out.println("Number of special Characters: "
            + count);
23
24         sc.close();
```

Enter the statement: Modi Birthday @ September 17, #&\$% is the
wishes code for him
Number of special Characters: 6

=== Code Execution Successful ===

BinaryConversion.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class BinaryConversion {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter Binary Number: ");
9         String binary = sc.nextLine();
10
11         // Check if input contains only 0 and 1
12         if (!binary.matches("[01]+")) {
13             System.out.println("Invalid Binary Number!");
14             sc.close();
15             return;
16         }
17
18         int decimal = Integer.parseInt(binary, 2);
19         String octal = Integer.toOctalString(decimal);
20
21         System.out.println("Decimal Number: " + decimal);
22         System.out.println("Octal: " + octal);
23
24         sc.close();
25     }
26 }
```

Enter Binary Number: 1101

Decimal Number: 13

Octal: 15

=== Code Execution Successful ===

NthLargestNumber.java



Share

Run

Output

Clear

```
1- import java.util.Arrays;
2 import java.util.Scanner;
3
4- public class NthLargestNumber {
5
6-     public static void main(String[] args) {
7
8         Scanner sc = new Scanner(System.in);
9
10        int[] arr = {14, 67, 48, 23, 5, 62};
11
12        System.out.print("Enter the value of N: ");
13
14        // Check whether input is an integer
15-        if (!sc.hasNextInt()) {
16            System.out.println("Invalid Input! Please enter
                an integer.");
17            sc.close();
18            return;
19        }
20
21        int n = sc.nextInt();
22
23-        if (n <= 0) {
24            System.out.println("Invalid Input! N should be
                greater than 0.");
```

Enter the value of N: 4

4th Largest number: 23

=== Code Execution Successful ===

RecursiveFactorial.java



Share

Run

Output

Clear

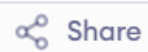
```
1 import java.util.Scanner;
2
3 public class RecursiveFactorial {
4
5     // Recursive function to calculate factorial
6     static long factorial(int n) {
7         if (n == 0 || n == 1)
8             return 1;
9         return n * factorial(n - 1);
10    }
11
12    public static void main(String[] args) {
13
14        Scanner sc = new Scanner(System.in);
15
16        System.out.print("Enter the value of n: ");
17
18        // Check for integer input
19        if (!sc.hasNextInt()) {
20            System.out.println("Invalid Input! Please enter
                an integer.");
21            sc.close();
22            return;
23        }
24    }
```

Enter the value of n: 6

The factorial of 6 is: 720

=== Code Execution Successful ===

CharacterCount.java



Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class CharacterCount {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int upperCount = 0;
9         int lowerCount = 0;
10        int digitCount = 0;
11
12        System.out.println("Enter * to exit...");
13
14        while (true) {
15
16            System.out.print("Enter any character: ");
17            char ch = sc.next().charAt(0);
18
19            if (ch == '*')
20                break;
21
22            if (Character.isUpperCase(ch))
23                upperCount++;
24            else if (Character.isLowerCase(ch))
25                lowerCount++;
26            else if (Character.isDigit(ch))
```

```
Enter * to exit...
Enter any character: W
Enter any character: d
Enter any character: A
Enter any character: G
Enter any character: g
Enter any character: H
Enter any character: *
Total count of lower case: 2
Total count of upper case: 4
Total count of numbers = 0

=== Code Execution Successful ===
```

AveragePositiveNegative.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class AveragePositiveNegative {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int num;
9         int positiveCount = 0, negativeCount = 0;
10        int positiveSum = 0, negativeSum = 0;
11
12        System.out.println("Enter -1 to exit...");
13
14        while (true) {
15
16            System.out.print("Enter the number: ");
17
18            if (!sc.hasNextInt()) {
19                System.out.println("Invalid Input! Please
20                    enter an integer.");
21                sc.close();
22                return;
23            }
24
25            num = sc.nextInt();
```

```
Enter -1 to exit...
Enter the number: 7
Enter the number: -2
Enter the number: 9
Enter the number: -8
Enter the number: -6
Enter the number: -4
Enter the number: 10
Enter the number: 1
Enter the number: -1
The average of negative numbers is: -5.0
The average of positive numbers is: 6.75

=== Code Execution Successful ===
```

StudentGrade.java



Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class StudentGrade {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         String[] subjects = {"Python", "C Programming",
9                               "Mathematics", "Physics"};
10        int[] marks = new int[4];
11        int total = 0;
12
13        for (int i = 0; i < 4; i++) {
14
15            System.out.print("Enter the marks in " +
16                             subjects[i] + ": ");
17
18            // Check for integer input
19            if (!sc.hasNextInt()) {
20                System.out.println("Invalid Input! Marks
21                                   should be integers between 0 and 100.");
22                sc.close();
23                return;
24            }
25
26            marks[i] = sc.nextInt();
27        }
28
29        // Calculate total marks
30        for (int i = 0; i < marks.length; i++) {
31            total += marks[i];
32        }
33
34        // Calculate aggregate marks
35        double aggregate = (double) total / marks.length;
36
37        // Print results
38        System.out.println("Total = " + total);
39        System.out.println("Aggregate = " + aggregate);
40
41        // Determine grade
42        if (aggregate >= 90) {
43            System.out.println("DISTINCTION");
44        } else {
45            System.out.println("FAIL");
46        }
47    }
48 }
```

```
Enter the marks in Python: 90
Enter the marks in C Programming: 91
Enter the marks in Mathematics: 92
Enter the marks in Physics: 93
Total = 366
Aggregate = 91.5
DISTINCTION

=== Code Execution Successful ===
```

PerfectNumbers.java



Share

Run

Output

Clear

```
39         System.out.println("Invalid Input! N should be
        greater than 0.");
40         sc.close();
41         return;
42     }
43
44     System.out.print("First " + n + " perfect numbers
        are: ");
45
46     int count = 0;
47     int number = 2;
48
49     while (count < n) {
50         if (isPerfect(number)) {
51             System.out.print(number);
52             count++;
53
54             if (count < n)
55                 System.out.print(", ");
56         }
57         number++;
58     }
59
60     sc.close();
61 }
62 }
```

```
Enter N: 3
First 3 perfect numbers are: 6, 28, 496
=== Code Execution Successful ===
```

EmployeeBonus.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class EmployeeBonus {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the grade of the employee: ")
9             );
10        String grade = sc.next().toUpperCase();
11
12        System.out.print("Enter the employee salary: ");
13
14        // Check if salary is numeric
15        if (!sc.hasNextDouble()) {
16            System.out.println("Invalid salary input.");
17            sc.close();
18            return;
19        }
20
21        double salary = sc.nextDouble();
22
23        // Check for negative salary
24        if (salary < 0) {
25            System.out.println("Salary cannot be negative.");
26        }
27    }
28 }
```

```
Enter the grade of the employee: B
Enter the employee salary: 50000
Salary = 50000.0
Bonus = 5000.0
Total to be paid: 55000.0
```

=== Code Execution Successful ===

DecimalConversion.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class DecimalConversion {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter Decimal Number: ");
10        String input = sc.nextLine();
11
12        try {
13            // Convert only if the input is a valid integer
14            int decimal = Integer.parseInt(input);
15
16            if (decimal < 0) {
17                System.out.println("Please enter a non
18                    -negative decimal number.");
19            } else {
20                String binary = Integer.toBinaryString
21                    (decimal);
22                String octal = Integer.toOctalString(decimal
23                    );
24
25                System.out.println("Binary Number = " +
26                    binary +
27                    "\nOctal = " + octal);
28            }
29        } catch (Exception e) {
30            System.out.println("Invalid input");
31        }
32    }
33 }
```

```
Enter Decimal Number: 15
Binary Number = 1111
Octal = 17

=== Code Execution Successful ===
```


PalindromeChoice.java



Share

Run

Output

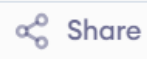
Clear

```
1- import java.util.Scanner;
2
3- public class PalindromeChoice {
4
5-     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.println("1. String Palindrome");
10        System.out.println("2. Number Palindrome");
11        System.out.print("Enter your choice: ");
12
13        int choice = sc.nextInt();
14        sc.nextLine(); // consume newline
15
16        switch (choice) {
17
18            case 1:
19                System.out.print("Enter the String: ");
20                String str = sc.nextLine();
21
22                String revStr = new StringBuilder(str
23                    ).reverse().toString();
24
25                if (str.equals(revStr))
```

```
1. String Palindrome
2. Number Palindrome
Enter your choice: 1
Enter the String: MADAM
Palindrome
```

=== Code Execution Successful ===

ATMBalance.java



Share

Run

Output

Clear

```
1 import java.util.HashSet;
2 import java.util.Scanner;
3
4 public class ATMBalance {
5
6     public static void main(String[] args) {
7
8         Scanner sc = new Scanner(System.in);
9
10        int totalBalance = 0;
11        HashSet<Integer> used = new HashSet<>();
12
13        int[] validDenominations = {2000, 500, 200, 100};
14
15        for (int i = 1; i <= 4; i++) {
16
17            System.out.print("Enter the " + i + "
18                           Denomination: ");
19            int denomination = sc.nextInt();
20
21            boolean valid = false;
22
23            // Check whether denomination is valid
24            for (int d : validDenominations) {
25                if (denomination == d) {
26                    valid = true;
```

```
Enter the 1 Denomination: 500
Enter the number of notes: 4
Enter the 2 Denomination: 100
Enter the number of notes: 20
Enter the 3 Denomination: 200
Enter the number of notes: 32
Enter the 4 Denomination: 2000
Enter the number of notes: 1

Total Available Balance in ATM: 12400
```

=== Code Execution Successful ===

MthMaxNthMin.java



Share

Run

Output

Clear

```
1 import java.util.Arrays;
2
3 public class MthMaxNthMin {
4
5     public static void findResult(int[] arr, int M, int N) {
6
7         // Check if M and N are valid
8         if (M <= 0 || N <= 0) {
9             System.out.println("Invalid Input! M and N
              should be greater than 0.");
10            return;
11        }
12
13        // Sort the array
14        Arrays.sort(arr);
15
16        int size = arr.length;
17
18        // Check if M and N are within range
19        if (M > size || N > size) {
20            System.out.println("Invalid Input! M or N
              exceeds array size.");
21            return;
22        }
23
24        int mthMax = arr[size - M];
```

```
1st Maximum Number = 89
3rd Minimum Number = 25
Sum = 114
Difference = 64
```

=== Code Execution Successful ===

PrimeCompositeCount.java



Share

Run

Output

Clear

```
1 import java.util.Scanner;
2
3 public class PrimeCompositeCount {
4
5     // Method to check whether a number is prime
6     public static boolean isPrime(int n) {
7         if (n <= 1)
8             return false;
9
10        for (int i = 2; i <= Math.sqrt(n); i++) {
11            if (n % i == 0)
12                return false;
13        }
14        return true;
15    }
16
17    public static void main(String[] args) {
18
19        Scanner sc = new Scanner(System.in);
20
21        int primeCount = 0;
22        int compositeCount = 0;
23
24        System.out.print("Enter the number of inputs: ");
25        int n = sc.nextInt();
26        sc.nextLine(); // consume newline
```

Enter the number of inputs: 8

Enter the numbers:

4

54

63

5

89

45

32

21

Composite number: 6

Prime number: 2

=== Code Execution Successful ===